

codeHive

Linked Lists

DYNAMIC MEMORY ARCHITECTURE

DSA Series: Data Structures Vol 01

1. The Linked Concept

A **Linked List** is a linear data structure where elements are not stored in contiguous memory locations. Instead, each element is a separate object called a **Node**. These nodes are "linked" using pointers, creating a flexible chain of data.

Anatomy of a Node

Every node consists of two essential parts:

- **Data:** The actual value you want to store (Integer, String, etc.).
- **Pointer (Next):** A reference that stores the memory address of the next node in the list.



2. Arrays vs. Linked Lists

Choosing between an array and a linked list depends on how you intend to access and modify your data.

Feature	Arrays	Linked Lists
Memory Storage	Contiguous Blocks	Scattered (Non-contiguous)
Size	Fixed (Static)	Dynamic (Grows/Shrinks)
Access Speed	$O(1)$ - Random Access	$O(n)$ - Sequential Access

Feature	Arrays	Linked Lists
Insertion/Deletion	Slow (Requires Shifting)	Fast ($O(1)$ at known position)
Memory Efficiency	Lower per element	Higher (Pointer overhead)

The Memory Advantage

In an array, if you want to insert a new element at the beginning, you have to move every other element one step to the right. In a **Linked List**, you simply create a new node and update its pointer to point to the current head. **No shifting required!**

3. Why Use Linked Lists?

1. Dynamic Size

You don't need to know the size of your list in advance. It can grow as long as there is free memory available in the system.

2. Optimized Memory Utilization

Arrays require a single large block of memory. If your memory is fragmented (small gaps of free space), an array might fail to allocate even if enough total space exists. Linked lists can utilize these small scattered gaps.

The Trade-off: No Random Access

In an array, you can jump to `arr[500]` instantly. In a Linked List, to find the 500th node, you must start at the **Head** and follow the pointers through 499 nodes. This is why search speed is slower ($O(n)$).

4. Knowledge Check

Exercise: Fill in the blanks.

Each node in a Linked List contains _____, and a _____ to where the next node is placed in memory.

Answer: Data | Pointer (Link)